

Data Platform Operating Model: Ownership, Governance, and Enterprise Accountability

Manuel Hernandez Giuliani

Independent Researcher and Instructor

manuelhernandezgiuliani.com

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Abstract

The maturity of contemporary data platforms depends increasingly less on isolated infrastructure and increasingly more on the operating model that organizes responsibilities, governance mechanisms, and value creation. In modern analytical ecosystems, the discussion is no longer limited to where information should be stored or processed, but rather to how ownership should be distributed, how quality should be governed, how costs should be managed, and how data products can sustain business, regulatory, and algorithmic use at scale. This article examines the operating model of data platforms from a socio-technical perspective, with particular emphasis on the paradigms of data as a product, platform as a product, domain ownership, and federated governance. It also analyzes platform operating layers, data contracts, service-level tiering, modern data and artificial intelligence governance, FinOps practices, and the Hub & Spoke enterprise governance model. The central argument is that a data platform can only be regarded as strategically mature when it integrates architecture, governance, distributed accountability, and financial discipline within a single organizational operating model (Dehghani, 2020; FinOps Foundation, 2026; SAP, n.d.; Tabassi, 2023).

Keywords: data platform, data as a product, data mesh, data fabric, data governance, FinOps, Hub & Spoke, data contracts

1. Introduction

Recent developments in data ecosystems have shown that an organization's analytical advantage does not depend only on the quality of its tools, but on the operating model that coordinates strategy, ownership, execution, and governance. Historically, centralized

approaches dominated by technology departments offered homogeneity and central governance, but they also generated bottlenecks, limited responsiveness, and a semantic gap between those who produce data and those who use it to operate or make decisions (Hernandez Giuliani, 2024a; IntechOpen, 2024).

In response to this limitation, organizational models have gained relevance that treat data as products and the analytical platform as an enabling product. This transition has been reinforced by the contemporary debate around data mesh, data fabric, and federated governance, all of which attempt to resolve the tension between domain autonomy, enterprise interoperability, and institutional governance. In this context, the platform ceases to be merely a collection of technical services and instead functions as an institutional architecture of responsibilities.

This article examines the operating model of data platforms from both an academic and applied perspective. To do so, it integrates governance-oriented sources and local teaching materials centered on the ManOxCo case in order to explain how ownership is distributed, how operating layers are structured, which governance mechanisms sustain the lifecycle of data products, and why financial governance and artificial intelligence governance have become inseparable components of platform design (Hernandez Giuliani, 2024a, 2024b, 2024c; IntechOpen, 2024).

2. Objectives

This article pursues four principal objectives. First, it analyzes the organizational paradigm that treats both data and the platform as governed and measurable products. Second, it examines the distribution of responsibilities across operating layers, business domains, and mechanisms of accountability. Third, it studies the governance instruments that make this model viable, particularly data contracts, service-level tiering, automated governance, and artificial intelligence risk management. Fourth, it evaluates how FinOps disciplines, service catalogs, and the Hub & Spoke model transform the data platform into an economically traceable and strategically sustainable enterprise capability.

3. Main Discussion

3.1. Organizational Paradigm, Ownership, and Accountability

A mature data platform requires a conceptual shift: data must be managed as products with defined consumers, explicit quality attributes, and unambiguous responsibilities for their maintenance. This idea, widely discussed in the data mesh conversation, implies that data assets are not merely technical byproducts, but value-bearing artifacts that must be discoverable, trustworthy, versionable, and operable under real business conditions (Dehghani, 2020; Hernandez Giuliani, 2024a). In the same way, the shared infrastructure is no longer conceived as a passive set of components, but as a platform as a product, that is, an internal service designed to reduce friction and enable self-service at scale (Hernandez Giuliani, 2024a; SAP, n.d.).

This shift requires a precise distinction among several dimensions of ownership. The first is data ownership, normally associated with the business domain that understands its semantics and operational context. The second is platform ownership, which belongs to the engineering function responsible for shared infrastructure, foundational security, and computational capacity. The third concerns ownership of the data product across its lifecycle, from design and adoption to obsolescence. The fourth concerns governance ownership, linked to risk, compliance, privacy, and auditability. When these responsibilities are not clearly distinguished, structural conflicts emerge that degrade quality, slow decision-making, and make accountability diffuse (Hernandez Giuliani, 2024a; SAP, n.d.).

3.2. Operating Layers of the Data Platform

The operating model becomes executable when the organization distinguishes layers with concrete functions. At the strategic layer, the Chief Data Officer, the data office, and platform product leaders define priorities, allocate investment, and connect analytical capability to corporate objectives. This layer does not operate pipelines or specific datasets; rather, its mission is to orient the platform, establish prioritization criteria, and prevent platform evolution from being subordinated to tactical urgencies disconnected from strategy (Hernandez Giuliani, 2024a; IntechOpen, 2024).

The platform layer provides the self-service environment and common services: infrastructure, storage, compute, security, observability, and automation. By contrast, the domain layer concentrates semantic and operational responsibility over data that represent real business processes, including cases in which IT systems and OT telemetry converge.

Finally, the data delivery layer materializes the technical flow through ELT pipelines, modeling, optimization, and patterns such as medallion architecture. The relevance of this scheme lies in the fact that it distributes work without fragmenting governance; each layer has relative autonomy, but also explicit obligations with respect to quality, reliability, and interoperability (Hernandez Giuliani, 2024a, 2024b, 2024c).

3.3. Data as a Product, Contracts, and Service Levels

If data must operate as products, then the relationship between producer and consumer cannot rest on informal agreements. At this point, data contracts emerge as one of the most important governance mechanisms in the operating model. A data contract specifies schemas, compatibility, versioning rules, quality metrics, access conditions, and service expectations. Its value lies in transforming integration into a verifiable agreement rather than an implicit dependency between teams (Hernandez Giuliani, 2024a; Hernandez Giuliani, 2024d).

This principle is reinforced by service-level objective tiering. Not all data products require the same operational criticality, and assuming otherwise leads to costly overengineering. For this reason, products are often classified into levels ranging from assets that are critical to life or core operations to analytical or experimental environments with much more limited guarantees. This stratification links criticality, cost, monitoring, and business sponsorship, and it requires the organization to justify economically any exceptional demand for availability, support, or recovery (Hernandez Giuliani, 2024a; Hernandez Giuliani, 2024d).

3.4. Modern Governance and Artificial Intelligence Governance

Contemporary governance can no longer depend exclusively on manual committees or late-stage documentary reviews. In modern platforms, governance is increasingly implemented as a system of automated governance controls that includes data lineage, active classification, policy enforcement, continuous monitoring, and dynamic access-control mechanisms. Within this framework, RBAC and ABAC are not merely security accessories, but structural components that ensure permissions, masking, and regulatory constraints are applied in contextual and verifiable ways (Hernandez Giuliani, 2024c; IntechOpen, 2024).

The expansion of artificial intelligence introduces an additional layer of complexity. Governing models and agents requires supervision of risk, model drift, explainability, decision traceability, and the relationships among training data, deployed models, and automated outputs. In the absence of these controls, an organization may scale analytical capability without scaling responsibility, thereby increasing regulatory exposure and

operational risk. For this reason, AI governance should be interpreted as a natural extension of data governance rather than as a separate or post-deployment discipline (Hernandez Giuliani, 2024b, 2024c; Tabassi, 2023).

Operationally, this governance should follow a continuous cycle of registration, policy assignment, pre-deployment review, runtime monitoring, and post-incident feedback. The enforcement layer is the platform control plane itself: identity policies, contract registries, lineage and catalog services, masking rules, and SLO monitoring. In a ManOxCo setting, that means governance is not a committee note added after deployment; it is the active mechanism that determines who can access a life-critical product, under which conditions a model or agent may use it, and how exceptions are escalated and audited (Hernandez Giuliani, 2024c; Tabassi, 2023).

3.5. FinOps, Metrics, and the Sustainability of the Operating Model

An operating model for a data platform is incomplete if it does not translate technical decisions into observable economic consequences. FinOps performs this function by introducing metrics and practices for assigning costs, making consumption transparent, and creating financial accountability around storage, compute, pipelines, domains, and data products. The showback model, and in some cases chargeback, makes it possible to turn the platform into an economically legible environment in which every demand for availability, latency, or complexity can be associated with an explicit cost (FinOps Foundation, 2026; Hernandez Giuliani, 2024a; IntechOpen, 2024).

This financial discipline is complemented by a stratified system of metrics. The platform measures SLA compliance, cost efficiency, and deployment frequency; the domain measures quality, adoption, and business impact; and the delivery layer measures lead time, pipeline reliability, and data freshness. What matters is not only measuring extensively, but measuring according to the actual responsibility of each actor. In this sense, metrics become a governance device because they connect accountability, prioritization, and budgetary legitimacy within the operating model (Hernandez Giuliani, 2024a, 2024d).

3.6. Hub & Spoke, Service Catalog, and Enterprise Governance

Complex organizations often resolve the tension between centralization and autonomy through hybrid models. The Hub & Spoke approach expresses this logic: the hub concentrates foundational assets that require corporate consistency, such as master data, regulatory datasets, or certified KPIs, while the spokes represent domains with the capacity to develop

data products, local analytics, and machine learning artifacts tailored to their specific needs. This pattern makes it possible to decentralize execution without losing the ability to standardize what the enterprise considers critical or normatively sensitive (Hernandez Giuliani, 2024a, 2024b; SAP, n.d.).

At the operating-model level, however, Hub & Spoke should not be treated as synonymous with either data mesh or data fabric. Data mesh primarily answers the question of ownership: it places semantic responsibility and product accountability in the business domains closest to the data. Hub & Spoke answers the question of governance distribution: it determines which cross-enterprise assets remain governed centrally and which decisions can be delegated to bounded domains. Data fabric, by contrast, is better understood in this article as a coordinating layer of metadata, lineage, policy automation, and cross-system discoverability that helps decentralized and hybrid organizations remain coherent without forcing all ownership back into a single central team. In that sense, mesh allocates accountability, Hub & Spoke allocates governance boundaries, and fabric improves integration and policy execution across both models (Dehghani, 2020; Hernandez Giuliani, 2024d; SAP, n.d.).

This distinction is especially relevant for ManOxCo. Plant operations, logistics, and hospital supply can operate as domain spokes with responsibility for the semantic quality and timeliness of their data products, while a central hub can retain governance authority over life-critical inventory definitions, certified service indicators, and regulatory policy. Fabric capabilities then provide the connective tissue, such as cataloging, lineage, and policy enforcement, that allows those domains to collaborate without surrendering all local accountability. The purpose of this bridge is therefore organizational rather than architectural: it clarifies how ownership, enterprise governance, and metadata-driven coordination can coexist without reopening the storage and reference-architecture discussion developed elsewhere (Hernandez Giuliani, 2024a, 2024d).

The practical viability of this model also depends on a clear catalog of platform services. Batch, streaming, CDC, storage, compute engines, data catalog, identity management, contract registry, and observability are not merely technical pieces; they constitute the operating portfolio that the platform offers to domains. When these services are coherent, documented, and reusable, the platform truly behaves as a product. When they are not, decentralization degenerates into fragmentation, duplication, and constant dependence on manual intervention (Hernandez Giuliani, 2024a, 2024d).

3.7. Operating Model Anti-Patterns

Recent literature on data mesh maturity and federated governance shows that failure is usually organizational before it is technological. A first anti-pattern consists in allowing IT to govern the meaning of data instead of the domain that produces it. A second anti-pattern appears when the language of domains or products is adopted without delegating real responsibility or decision-making capacity. A third arises when governance is designed as reactive bureaucracy rather than as an automated system of governance controls. Finally, the adoption of AI without traceability, explainability, or drift management represents one of the most costly errors of the new analytical environment (Dehghani, 2020; SAP, n.d.; Tabassi, 2023).

These anti-patterns matter because they reveal a deeper methodological lesson: it is not enough to introduce new organizational terms if the underlying structures of power, responsibility, and measurement remain intact. The transformation of the operating model becomes real only when the organization changes the way it assigns ownership, designs contracts, monitors risk, and justifies economically the operation of its data fabric.

4. Conclusion

The operating model of data platforms now constitutes a strategic dimension of enterprise architecture. Its function is not only to organize teams or describe technical layers, but to establish a stable relationship among distributed responsibility, institutional governance, self-service capability, and financial sustainability. In this sense, the transition toward data as a product and platform as a product should be read as a large-scale organizational transformation rather than as a simple terminological preference (Dehghani, 2020; SAP, n.d.).

The evidence reviewed suggests that platform maturity depends on the articulation of several components: explicit ownership, data contracts, tiered service levels, automated governance, artificial intelligence risk management, and FinOps discipline. The value of the model lies precisely in that integration. When these elements are designed coherently, the organization gains speed, traceability, and autonomy; when they are fragmented, the platform becomes costly, opaque, and difficult to sustain (FinOps Foundation, 2026; Hernandez Giuliani, 2024a, 2024c; Tabassi, 2023).

5. Discussion Highlights

The discussion supports four principal findings. First, the modern data platform should be understood as an organizational operating model rather than only as technical infrastructure. Second, decentralization produces value only when it is accompanied by explicit mechanisms of ownership, contracts, and federated governance. Third, FinOps and artificial intelligence governance are no longer peripheral domains, but structural components of analytical operations. Fourth, data mesh, Hub & Spoke, and data fabric should not be treated as interchangeable labels, because they address different operating-model questions concerning accountability, governance boundaries, and cross-enterprise coordination.

6. Glossary

Data contract: A verifiable agreement between producer and consumer that defines schema, versioning, quality, and service expectations.

Data as a product: An approach that treats data assets as artifacts with consumers, metrics, lifecycle, and explicit responsibility.

Data fabric: A coordination approach that uses metadata, lineage, and policy automation to unify governance across distributed data environments.

Data mesh: A socio-technical operating model that decentralizes data ownership to business domains and treats data as a product.

FinOps: An operating discipline that links technology consumption and financial accountability in order to optimize platform costs.

Hub & Spoke: A hybrid organizational model that combines a central core of enterprise assets with decentralized domains.

Data observability: The ability to monitor the health, reliability, freshness, and operational behavior of data and pipelines.

Platform as a product: The conception of the analytical platform as a reusable, self-service internal service oriented toward internal users.

SLO tiering: The stratification of service-level objectives according to operational criticality and business cost.

7. References

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Appendix A. Self-Assessment Exercises

1. What main difference distinguishes a data-as-a-product approach from a purely traditional technical management model?

- A. It eliminates the need for domain owners.
- B. It treats data as assets with consumers, metrics, lifecycle, and explicit responsibility.
- C. It replaces governance with financial dashboards.
- D. It restricts the use of data contracts to experimental environments.

2. What is the principal function of a data contract within the operating model?

- A. To replace observability with manual reviews.
- B. To define schema, versioning, quality, and service conditions between producer and consumer in a verifiable way.
- C. To assign business semantics to the infrastructure team.
- D. To convert every data product into a Tier 0 asset.

3. Which statement best describes the logic of the Hub & Spoke model?

- A. It centralizes all data and prohibits domain analytical autonomy.
- B. It eliminates master data in favor of total flexibility.
- C. It combines a central core of enterprise assets with decentralized domains capable of developing their own products.
- D. It delegates security exclusively to external providers.

4. Why is FinOps part of the operating model rather than merely a technology accounting practice?

- A. Because it connects technical decisions with consumption, costs, and financial accountability.
- B. Because it replaces data quality metrics.
- C. Because it prevents the use of cloud storage.
- D. Because it eliminates the need for strategic prioritization.

5. Which of the following belongs most clearly to artificial intelligence governance?

- A. Only the creation of relational schemas.
- B. Model drift, explainability, and traceability of automated decisions.
- C. The calculation of cost per terabyte.
- D. The definition of certified hub KPIs.

6. Which anti-pattern most directly compromises accountability in the operating model?

- A. The business domain assuming semantic responsibility for the data.
- B. The existence of reusable service catalogs.
- C. IT defining the meaning of the data instead of the responsible domain.
- D. The classification of data products by operational criticality.

Appendix B. Answer Key

1. **B.** Data is managed as a product with consumers, metrics, lifecycle, and explicit responsibility.
2. **B.** The data contract formalizes schema, versioning, quality, and service expectations.
3. **C.** Hub & Spoke combines central control over foundational assets with domain autonomy.

4. **A.** FinOps introduces financial traceability and connects cost to architecture and operating decisions.
5. **B.** AI governance includes model risk, drift, explainability, and traceability.
6. **C.** When IT controls the meaning of data, domain responsibility becomes blurred.

Appendix C. Citation Mapping and Notes on the Source Base

- **Introduction:** Dehghani (2020), Hernandez Giuliani (2024a, 2024b, 2024c), IntechOpen (2024), SAP (n.d.), and Tabassi (2023).
- **Section 3.1:** Dehghani (2020), Hernandez Giuliani (2024a), and SAP (n.d.).
- **Section 3.2:** Hernandez Giuliani (2024a, 2024b, 2024c) and IntechOpen (2024).
- **Section 3.3:** Hernandez Giuliani (2024a, 2024d).
- **Section 3.4:** Hernandez Giuliani (2024b, 2024c), IntechOpen (2024), and Tabassi (2023).
- **Section 3.5:** FinOps Foundation (2026), Hernandez Giuliani (2024a, 2024d), and IntechOpen (2024).
- **Section 3.6:** Dehghani (2020), Hernandez Giuliani (2024a, 2024b, 2024d), and SAP (n.d.).
- **Section 3.7:** Dehghani (2020), SAP (n.d.), and Tabassi (2023).